TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY

TEAMG4 · PRESENTATION 1 · THE SOLUTION PITCH

SaigonBite

Cloud-Kitchen Food Delivery Hub for Ho Chi Minh City

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Software Engineering (SE) · Semester 1, 2026-2027 · Sem01_2627 · Tier 4

01 THE PROBLEM **02** OUR SOLUTION **03** OWNERSHIP **04** DE-RISKING

Customers and kitchen load

Clear prices and controlled delivery for local customers and foreign tourists.

2

customer groups, one lifecycle: a ten-digit mobile, or a passport and a hotel room.

5

payment methods, three per group; only cash on delivery appears in both lists (BR13).

18

business rules define pricing, payment, lifecycle and registration (BR1–BR18).

Quiet Kitchen	< 30 %		10 % off the food subtotal (BR7)
Normal	30–<90 %		no Quiet Kitchen discount
Refused	≥ 90 %		new orders refused (BR8)

Load uses Confirmed + Preparing orders / capacity. **29.5 % qualifies** for the discount; **90 % refuses** a new order (BR7, BR8).

A layered, testable solution

Pillar 01

Load-aware pricing

Nine calculation steps; round once at the end. Business services check item availability and hub membership.

Pillar 02

A payment Strategy

One interface, one class per method and one registration; existing payment classes stay unchanged (NFR2).

Pillar 03

An enforced lifecycle

Eight states, eight transitions, named actors. The business layer rejects the rest (BR15).

Presentation	Razor views · Bootstrap 5 · Chart.js · English / Vietnamese resources
↓	
Business	**PricingEngine** + validation · **OrderStateMachine** BR15–BR16 · **IPaymentStrategy** + resolver
↓	
Data	EF Core 8 Code-First · SQL Server 2022 · soft-delete filters · price snapshot

Planned isolation: pricing and state rules can be unit-tested without a database or browser. Verify in Part 3.

Three owners, complete CRUD scopes

MEMBER	MAIN USE CASE — COMPLETE CRUD	SECONDARY SCOPE	REPORT OWNED
Ngô Lâm Tiến 524H0195	**D — Place and Manage Orders** **MANDATORY** · pricing and payment Strategy	C — Manage Vouchers	—
Nguyễn Công Thành 524H0191	**A — Manage Menu & Brands** brands, items, availability, soft delete	E — Manage User Accounts	R1 — Revenue
Nguyễn Trọng Phú 524H0179	**B — Manage Hubs & Capacity** hubs, capacity, derived load	F — Manage Deliveries Change-request note	R2 — Hub Utilization

> **Each member owns one main use case with complete CRUD.** R1 aggregates brand category and payment method, so it sits with A. R2 aggregates load and preparation time, so it sits with B. D carries the pricing engine and both customer types, so its owner takes no report.

Acceptance targets and open decisions

4

Tier 4 · cap 10.0/10: Tier 3 + live sandbox API + interactive Admin dashboard + automated tests. Core MVC variant pending confirmation.

20+

Planned EP/BVA cases: BR1–BR12 and BR14, plus at least five forbidden-transition tests.

1

New payment class + one registration (NFR2). Assess other change requests separately.

AMBIGUITY A1

Who assigns the shipper?

An Admin assigns an available shipper to a Ready order; store the shipper and assignment time. No automatic dispatch.

Working assumption; instructor review pending.

AMBIGUITY A2

What is refunded after a failed delivery?

Refund online payments in full (simulated); COD collects nothing. Failed is terminal and excluded from revenue.

Working assumption; instructor review pending.